ANA-MARIA LUISA MOCANU

Data Scientist & NLP Researcher

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EXPERIENCE

Data Scientist

Endava

Jun 2026 – present · Bucharest, Romania

- Build ML models on card-payment data, such as fraud and risk scoring and authorization-rate optimization.
- Explore, clean and analyse transaction data (Python, SQL), engineer features and present findings to clients.
- Build LLM-based solutions and deploy AI agents on Google Cloud (Gemini Enterprise).

Software Engineer

Endava

Dec 2021 – Jun 2026 · Bucharest, Romania

- Built and maintained fintech applications from scratch in Java and Spring Boot for clients including a top-3 UK bank, a premium German carmaker's payments system and a European e-invoicing platform.
- Grew from independent feature work into owning test automation and DevOps: deployments, Kong gateway configuration and release preparation, working closely with clients and product owners.
- Delivered features in service-oriented and microservice architectures in an Agile team; one project was nominated among Endava's most successful.
- Technologies: Java, Spring Boot, Angular, JavaScript, PostgreSQL, Cosmos DB, MongoDB, Redis, Liquibase, Mongock, Kafka, Docker, AWS, Azure, Maven, GitHub, Bitbucket, Jira, Bruno, Linux.

Python Teaching Assistant

University Politehnica Bucharest

Oct 2025 – present · Bucharest, Romania

- Teach Python to Bachelor's students in Google Colaboratory; design the lab exercises, homework and tests.
- Received 12 positive student reviews for making programming concepts easy to understand.
- Co-coordinate 2 Bachelor's and 2 Master's students on their thesis projects.
- Developed Scholar Rank, an auto-grading exam platform used for live lab exams.

RESEARCH & PUBLICATIONS

IROH: Insightful Ranking of Humor using Multi-Stage Hybrid Retrieval with Rationale-Distilled LLM Judges for JOKER 2026 Track Task 1 English

A.-M. L. Mocanu, S. Mocanu, C.-O. Truică, E.-S. Apostol
CLEF 2026 (oral) – JOKER Task 1: ***1st place*** *(MAP 0.6347)*
CEUR PDF · arXiv · Website · Code · Models

- Three-stage pipeline: hybrid BM25 + dense retrieval, a fine-tuned cross-encoder reranker, and an ensemble of LoRA/QLoRA-adapted LLM judges.

EDUCATION

PhD in Computer Science (NLP)

University Politehnica Bucharest

Oct 2025 – present

Thesis: Methods and Resources for Low-Resource Languages

Master in Artificial Intelligence

University Politehnica Bucharest

Oct 2021 – Sep 2025

Thesis: Aspect-Based Sentiment Analysis for Romanian Product Reviews

Bachelor in Mechanical Engineering

University Politehnica Bucharest

Oct 2017 – Jul 2021

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, JavaScript, SQL, Bash.
- **ML & NLP:** PyTorch, TensorFlow, Keras, Hugging Face Transformers, LoRA/QLoRA, LLM distillation, retrieval and reranking (BM25, dense, cross-encoders), CLIP, scikit-learn, LightGBM, spaCy, OpenCV, Pandas, NumPy, Matplotlib.
- **Backend & Web:** Spring Boot, FastAPI, Flask, Django, Angular, HTML, CSS.
- **Databases & Data:** PostgreSQL, MySQL, SQLite, Azure Cosmos DB, MongoDB, Redis, Kafka.
- **Cloud & DevOps:** Docker, Kubernetes (basic), CI/CD, Terraform, AWS, Azure, Google Cloud (Gemini Enterprise), Linux, Kong.
- **Tools & Collaboration:** Git, GitHub, Bitbucket, Jira, Maven, Gradle, Liquibase, Agile.

LANGUAGES

- Romanian (native), English (B2, professional).

PERSONAL SKILLS

- Critical Thinker, Adaptability, Good Interpersonal Skills, Time Management, Team Player, Problem Solver.

HOBBIES

- Learning new things, Sudoku, Listening to Music, Hiking.

- Trained the judges on query-aware rationales distilled from a vision-language model, plus structured hard negatives that teach the reranker to tell real jokes from lookalikes.

Through the Eyes of the Beholder: Biometric and Demographic Conditioning for Multimodal Sexism Detection

A.-M. L. Mocanu, S. Mocanu, C.-O. Truică, E.-S. Apostol
CLEF 2026 (oral and poster) – EXIST Subtask 2.2: top 26%
CEUR PDF · arXiv · Website · Code · Model

- Human-centered model (CLIP + XLM-RoBERTa) fusing meme text, a generated visual description, the image, annotator demographics and physiological signals; cross-attention grounds the description in the image, FiLM conditions on annotators.
- Framed as label distribution learning with a soft-label KL divergence loss and a supervised contrastive objective.

ACAT: A Collaborative Platform for Efficient Aspect-Based Sentiment Dataset Annotation

A.-M. L. Mocanu, C.-O. Truică, E.-S. Apostol
DaWaK 2026 (oral) – arXiv · Website

- Web platform (Flask, PostgreSQL, Docker) for four ABSA tasks of increasing granularity; computes Cohen's and Fleiss' kappa at export and exports to CSV, JSON and XML. Currently used by 400+ users.

Romanian Review (RoR): ABSA Dataset & LLM Distillation

- Created the first Romanian ABSA dataset with mixed aspects and multiple sentiments: 30,000+ e-commerce reviews across 14 product aspects, 3,000+ labeled manually and the rest by LLMs (GPT-4.1, Claude Haiku).
- Distilled the LLM annotations into ∼100M-parameter Romanian transformers (RoBERT, BERT Romanian); GPT-4.1-supervised students reached F1 0.787 and 0.788, matching their teacher at a fraction of the size.

PROJECTS

Fraud Detection Service

- Cleaned a raw 41.6K-row card-transaction dataset (mixed date formats, label spellings and currencies, duplicates, sentinel values, a leaking column) and split it by time to prevent leakage.
- Compared Random Forest + SMOTE, LightGBM and an LSTM at a 1.4% fraud rate; LightGBM reached PR-AUC 0.65 on the held-out month (73% precision at 61% recall).
- Served the model with FastAPI (input validation, uniform error responses, rate limiting) plus an AI agent (Strands Agents) that calls it as a tool and keeps conversation memory across stateless calls.
  Link: github.com/medz23/transaction-fraud-detection

Scholar Rank: Python Exam IDE

- Browser exam platform (Monaco editor) that grades student code in sandboxed Docker containers with memory and CPU limits, supporting function-based and stdin/stdout grading; used for live lab exams.
- Professor dashboard (problems, groups, test cases, randomized problem pools, CSV export) and anti-cheat measures (fullscreen enforcement, tab-switch strikes, disabled copy/paste).
  Technologies used: Python, Flask, PostgreSQL, Docker, JavaScript.
  Link: github.com/medz23/scholar-rank

Abusive Language Detection (2022)

- Trained 7 models (GloVe and one-hot CNNs; BERT, RoBERTa, BERTweet, HateBERT, XLNet) on 5 hate-speech datasets (Davidson, Founta, Waseem, Vidgen, OLID), combined with soft and hard voting.
  Link: github.com/brittleru/abusive-language-detection

CERTIFICATIONS

- Certified Partner Specialist – Gemini Enterprise Deployment – Google Cloud
- Gemini Enterprise Agent Development – Google Cloud – 2026 (completed, certificate pending)
- Claude Code in Action – Anthropic – Jul 2026
- Building with the Claude API – Anthropic – Jun 2026
- Introduction to Model Context Protocol – Anthropic
- Introduction to Agent Skills – Anthropic
- Applied Machine Learning – Google Digital Stack – Jun 2023
- NLP with Probabilistic Models – DeepLearning.AI – Jan 2022
- NLP with Classification and Vector Spaces – DeepLearning.AI – Jan 2022
- Android Fundamentals – Google Digital Stack – Jul 2021
- Python – Google Digital Stack – Feb 2021

PRESENTATIONS

- Oral presentations (JOKER, EXIST) and Poster (EXIST) – The Conference and Labs of the Evaluation Forum (CLEF) – Sep 2026
- Oral presentation – The 28th International Conference on Big Data Analytics and Knowledge Discovery (DaWaK) – Aug 2026
- Poster Session – Best Romanian AI Thesis Awards – Romanian AI Days – Sep 2026
- Poster Session – Romanian AI Days – Sep 2025
- Participated in NitroNLP hackathon – 2023

VOLUNTEERING

- Environmental projects – 'Act for Tomorrow' (2019)
- Community building – 'TEAM: Together Everyone Achieves More' (2014–2016)
- Humanitarian projects – 'Caravana Voluntarilor' (2014–2015)